

Algorithm & Programming Cheat Sheet

What is an Algorithm?

An algorithm is a clear, logical sequence of steps to solve a problem.

Think of it as a recipe!

Example:

1. Start
2. Get two numbers
3. Add them together
4. Show the result
5. Finish

Programming vs Program

- Program: The final code the computer runs.
- Programming: The process of writing that code from an algorithm.

C++ vs Python

Feature	C++	Python
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Type	Compiled	Interpreted
Syntax	Strict (lots of ;)	Simple (like English)
Speed	Fast	Slower
Difficulty	Medium-hard	Beginner-friendly
Use Case	Games, Systems	Data, AI, Web
Memory	Manual	Auto

Compiled vs Interpreted

- Compiled (C++): Translate whole code first, then run fast.
- Interpreted (Python): Run line by line, easier for testing.

Pointers & Linked Lists

C++:

- Uses pointers (memory addresses) to link data manually.
- Example: `Node* next;`

Python:

- No manual pointers.
- Uses objects to link data (Python handles the rest).
- Example:

```
class Node:
```

```
    def __init__(self, data):
```

```
        self.data = data
```

```
        self.next = None
```

Why? Python focuses on safety and simplicity.